Infrastructure is now digital

Bridges, water systems, and power grids rely on digital control systems. If those systems fail, the structure cannot function properly.

THE PROBLEM

Cyberattacks on infrastructure are increasing. In 2021, the Colonial Pipeline was shut down by a ransomware attack. This caused fuel shortages across 17 states and widespread panic buying. This shows how digital attacks can create real-world disruption — without touching a single physical component.

WHAT ENGINEERS CAN DO

INCLUDE CYBERSECURITY IN PROJECT PLANNING

WORK WITH IT AND CYBERSECURITY PROFESSIONALS

PERFORM REGULAR SYSTEM RISK ASSESSMENTS

TRAIN STAFF TO RECOGNIZE THREATS

WHY THIS MATTERS

CIVIL ENGINEERS FOCUS ON PHYSICAL SAFETY. BUT NOW CYBERSECURITY ALSO AFFECTS THAT SAFETY. IF TRAFFIC SYSTEMS, WATER PLANTS, OR POWER GRIDS ARE HACKED, PUBLIC SAFETY IS AT RISK. CYBERSECURITY MUST BE TREATED LIKE ANY OTHER DESIGN REQUIREMENT — BUILT IN FROM THE START, NOT ADDED LATER.

- TRAFFIC SYSTEMS CAUSING ACCIDENTS
- WATER PLANTS DISTRIBUTING UNSAFE WATER
- POWER GRIDS FAILING DURING EMERGENCIES
- EMERGENCY RESPONSE GOING OFFLINE